R2. The Returns Fraud vs. Genuine Grievance Sorter

Real-world pain: Online fashion returns run 25-40%. A small fraction are abusive (wardrobing, empty-box, serial returners), but blanket-blocking customers destroys loyalty. Retailers need to separate the abusive few from the genuinely unhappy many — and be able to explain the decision.

The problem: Score each incoming return request as `auto-approve`, `inspect`, or `escalate`, using both structured order history and the free-text reason the customer typed.

What we provide:

- 60k synthetic return requests: order value, category, discount used, delivery time, customer tenure, prior return rate, payment method
- A free-text `customer_reason` field (multilingual mix: English, Hinglish, transliterated Hindi) — deliberately messy, with typos
- Product catalogue with size charts and category
- Hidden labels: `abusive`, `genuine_quality_issue`, `genuine_fit_issue`, `genuine_wrong_item`

Assumptions students may make:

- No images are available (so no photo verification)
- Customer text may be honest, exaggerated, or fabricated — text alone is not sufficient evidence
- A false accusation costs far more than a missed fraud: teams must pick and defend a cost ratio (we suggest 1 wrong block ≈ 10 missed frauds)
- Returns policy: 30 days, free returns, no restocking fee

Desired output:

1. A three-way routing decision with a calibrated confidence score
2. A **human-readable justification** per decision, citing the specific signals used ("4th return of same size in 60 days; reason text matches prior submissions")
3. `evaluate.py` score = cost-weighted accuracy using the published cost matrix, not raw accuracy
4. A fairness check: show that the model does not disproportionately flag new customers, low-value orders, or non-English text — this is a scored requirement
5. A "size-chart fix" insight: which 5 products cause the most fit-related returns and should have their size guidance corrected

Stretch: a suggested resolution message to the customer, tone-matched to the reason category.